

Student Segmentation and Personalized Course Recommendation System Using K-Means Clustering and Machine Learning

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Abstract—In the era of digital learning platforms, the exponential growth of online educational content has created a pressing need for intelligent systems that can segment learners and deliver personalized course recommendations. This paper presents an end-to-end Student Segmentation and Recommendation System that leverages K-Means clustering, Principal Component Analysis (PCA), and behavioral feature engineering to identify distinct learner personas from transactional e-learning data. The system processes three interlinked datasets — Users, Courses, and Transactions — to derive nine rich behavioral features per learner, including total courses enrolled, diversity score, learning depth, average spending, and enrollment frequency. Unsupervised clustering with $k=4$ segments learners into four meaningful groups, validated by a silhouette score metric. A content-popularity-based recommendation engine then suggests top-5 courses for any given learner based on their cluster peers' behavior. The system is deployed as an interactive Streamlit web application, enabling real-time data upload, visual exploration, and personalized recommendations. Experimental results demonstrate that the proposed approach achieves a meaningful silhouette score and produces interpretable learner segments that align with real-world educational behavior patterns.

Index Terms—E-learning, K-Means clustering, learner segmentation, personalized recommendation system, machine learning, Principal Component Analysis, Streamlit, behavioral analytics.

I. INTRODUCTION

The rapid proliferation of Massive Open Online Courses (MOOCs) and e-learning platforms has transformed modern education. Platforms such as Coursera, Udemy, and edX host thousands of courses catering to diverse audiences. Despite the richness of available content, learners often face decision fatigue when choosing courses that align with their interests, skill level, and learning goals. Personalized recommendation systems have emerged as a critical component to address this challenge [1].

Traditional recommendation approaches such as collaborative filtering and content-based filtering rely heavily on explicit user ratings and item metadata. However, in educational contexts, learner behavior — including enrollment frequency, spending patterns, category diversity, and progression from beginner to advanced topics — provides richer signals for understanding learning intent and grouping similar learners [2].

This paper proposes a Student Segmentation and Recommendation System (SSRS) that combines unsupervised machine learning and behavioral feature engineering. The system is built using Python and deployed as an interactive Streamlit dashboard. It processes multi-sheet Excel data from an e-learning platform (EduPro), engineers nine behavioral features per learner, and applies K-Means clustering with PCA-based visualization to identify four distinct learner segments. A cluster-based recommendation engine then leverages peer behavior within each segment to surface the top five most popular courses for any given learner.

The key contributions of this work are:

- A nine-dimensional behavioral feature engineering pipeline derived from transactional e-learning data.

- Application of K-Means clustering ($k=4$) with silhouette score validation for learner segmentation.
- A cluster-based course recommendation engine requiring no explicit ratings or content metadata.
- An interactive Streamlit dashboard providing real-time segmentation, visualization, and recommendations.
- Empirical validation using the Elbow Curve and Silhouette Score metrics.

II. RELATED WORK

Learner modeling and segmentation in educational technology have attracted significant research attention. Romero and Ventura [3] provided a comprehensive survey of educational data mining techniques, highlighting clustering as a primary tool for understanding student behavior. Their work established that grouping learners by engagement patterns significantly improves personalization outcomes.

Perez-Alvarez et al. [4] applied K-Means clustering to MOOC participant data from Coursera and identified five distinct learner archetypes based on video-watching behavior and quiz performance. Their findings validated the utility of clustering for adaptive learning path generation. Similarly, Chaplot et al. [5] demonstrated that behavioral sequences in online courses could be used to predict learner dropout with over 80% accuracy, underscoring the predictive power of transactional engagement data.

Collaborative filtering remains the dominant paradigm for course recommendation [6]. However, its cold-start problem — the inability to make recommendations for new users or courses — limits applicability in growing platforms. Hybrid models combining collaborative filtering with content-based signals have been proposed [7], but they require substantial metadata and rating density. The cluster-based popularity approach used in this paper circumvents cold-start issues by relying solely on enrollment behavior within peer groups.

Deep learning approaches, including autoencoders and recurrent neural networks, have been applied to learner modeling with promising results [8]. However, their interpretability limitations and high computational requirements make them less suitable for real-time interactive dashboards intended for educational administrators. Lightweight unsupervised methods such as K-Means retain full interpretability and can be deployed efficiently in web applications.

PCA for dimensionality reduction and 2D visualization of learner clusters has been employed in prior studies [9]. Its use in this system follows established best practices for making high-dimensional behavioral spaces comprehensible to non-technical stakeholders.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Dataset Description

The system operates on the EduPro dataset, an Excel workbook comprising three interlinked sheets:

- **Users:** Contains UserID, Age, and Gender fields for each registered learner.
- **Courses:** Contains CourseID, CourseCategory, CourseType, CourseLevel (Beginner / Intermediate / Advanced), and CourseRating.
- **Transactions:** Contains UserID, CourseID, TransactionDate, and Amount (enrollment fee paid).

The three sheets are merged on their respective key columns (UserID and CourseID) to produce a unified transactional record for each enrollment event. Only learners with three or more course enrollments are retained to ensure sufficient behavioral signal for meaningful clustering.

B. Feature Engineering

Nine behavioral features are engineered per learner from the merged dataset. Table I summarizes these features.

TABLE I. LEARNER BEHAVIORAL FEATURES

Feature	Description
TotalCourses	Total courses enrolled by the learner

<i>AvgCoursesPerCategory</i>	Average enrollments per course category
<i>EnrollmentFrequency</i>	Unique transaction dates (activity days)
<i>PreferredCategory</i>	Most frequently enrolled course category
<i>PreferredLevel</i>	Most frequently enrolled course level
<i>AvgRating</i>	Mean rating of enrolled courses
<i>AvgSpending</i>	Mean transaction amount per enrollment
<i>DiversityScore</i>	Number of unique course categories explored
<i>LearningDepth</i>	Ratio of Advanced to (Beginner + 1) enrollments

Categorical features (*PreferredCategory* and *PreferredLevel*) are label-encoded using scikit-learn's *LabelEncoder*. All nine features are then standardized using *StandardScaler* to zero mean and unit variance, ensuring that features with larger numeric ranges do not dominate the distance calculations in K-Means.

C. K-Means Clustering

K-Means clustering partitions the learner feature space into k non-overlapping clusters by minimizing the within-cluster sum of squared distances (inertia). The algorithm iterates between assigning each learner to the nearest centroid and recomputing centroids until convergence. The objective function is:

$$J = \sum_i \sum_x C_i ||x - \mu_i||^2 \quad (1)$$

where μ_i is the centroid of cluster C_i and x is a learner feature vector. The optimal number of clusters $k=4$ was determined by applying the Elbow Method over the range $k=2$ to $k=9$, identifying the point of diminishing return in inertia reduction. The silhouette score was computed to quantitatively validate cluster quality.

The KMeans implementation from scikit-learn is used with `n_init='auto'` and `random_state=42` for reproducibility. The resulting cluster labels are appended to the learner profile dataframe.

D. Dimensionality Reduction with PCA

To enable 2D visual inspection of the nine-dimensional learner space, Principal Component Analysis is applied. PCA decomposes the standardized feature matrix X into orthogonal principal components ordered by explained variance. The first two principal components (PC1 and PC2) are retained and stored as *PCA1* and *PCA2* columns in the learner profiles dataframe. These are used to render the cluster scatter plot in the Streamlit dashboard.

E. Recommendation Engine

The recommendation engine follows a cluster-based collaborative filtering paradigm. For a target learner:

- Identify the learner's cluster assignment from the K-Means model.
- Retrieve all learners belonging to the same cluster.
- Aggregate all course enrollments of cluster peers from the Transactions dataset.
- Rank courses by enrollment frequency (popularity) within the peer group.
- Return the top 5 most popular courses as recommendations, along with their *CourseCategory* and *CourseLevel* metadata.

This approach requires no explicit ratings and naturally surfaces courses that learners with similar behavioral patterns have gravitated toward, making it robust to sparse rating data.

F. Streamlit Dashboard

The complete pipeline is deployed as a five-tab Streamlit web application. The Overview tab presents aggregate metrics (total learners, transactions, silhouette score), a course category pie chart, and a category-wise spending bar chart. The Cluster Dashboard tab renders the PCA scatter plot and a per-cluster summary statistics table. The Learner Explorer tab enables drill-down into individual learner profiles and transaction histories with a spending timeline chart. The Recommendations tab surfaces the top-5 peer-cluster course recommendations for a selected learner. The Model Validation tab

presents the Elbow Curve, silhouette score metric, and a feature correlation heatmap.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Dataset Statistics

The EduPro dataset used in experiments contains records across the Users, Courses, and Transactions sheets. After merging and filtering for learners with at least three enrollments, the pipeline produces a final learner profile matrix. The system processes this matrix end-to-end in under a few seconds on a standard laptop, demonstrating its practical efficiency for real-time interactive use.

B. Elbow Curve Analysis

The Elbow Curve, plotting inertia against the number of clusters k from 2 to 9, exhibits a characteristic inflection point at $k=4$. Beyond this value, reductions in inertia are marginal, confirming that four clusters provide an optimal balance between compactness and generalizability. This finding is consistent with prior studies on learner segmentation in MOOC environments [4].

C. Silhouette Score Validation

The silhouette score, ranging from -1 to +1, measures how similar each learner is to its own cluster compared to neighboring clusters. A score close to +1 indicates dense, well-separated clusters. The proposed system achieves a silhouette score in a range that indicates meaningful cluster separation, confirming that the four identified segments are distinct and interpretable rather than arbitrary partitions of the feature space.

D. Learner Segment Interpretation

Cluster analysis of the mean feature values per segment reveals four interpretable learner personas:

- Cluster 0 – Casual Explorers: Low *TotalCourses*, high *DiversityScore*, low *LearningDepth*. These learners sample many categories shallowly without deep specialization.
- Cluster 1 – Deep Specialists: High *LearningDepth*, low *DiversityScore*, high *AvgSpending*. These learners invest heavily in advancing within a single domain.
- Cluster 2 – Frequent Learners: High *EnrollmentFrequency*, moderate *TotalCourses*, moderate *DiversityScore*. These are highly active learners with broad but consistent engagement.
- Cluster 3 – Budget Beginners: Low *AvgSpending*, high *Beginner-level enrollment ratio*, moderate *TotalCourses*. These learners prioritize low-cost introductory content.

These personas align intuitively with observed behavior patterns in large-scale e-learning platforms and can directly inform targeted marketing, curriculum design, and scholarship allocation decisions by platform administrators.

E. Recommendation Quality

Recommendations generated by the cluster-based engine surface courses consistently aligned with the target learner's preferred category and level, as inferred from their cluster's aggregate behavior. While a formal precision-recall evaluation would require a held-out ground truth dataset — an avenue left for future work — the qualitative alignment of recommendations with learner profiles demonstrates the practical utility of the approach.

F. Feature Correlation Analysis

The heatmap of Pearson correlations among the nine features reveals strong positive correlation between *TotalCourses* and *EnrollmentFrequency*, and moderate correlation between *DiversityScore* and *AvgCoursesPerCategory*. *LearningDepth* exhibits low correlation with spending features, suggesting that depth of learning progression is relatively independent of financial investment. These correlations provide insight into the underlying structure of learner behavior and can guide future feature selection.

V. SYSTEM IMPLEMENTATION

The system is implemented entirely in Python 3.10+. The core ML pipeline uses pandas for data manipulation, scikit-learn for preprocessing (*StandardScaler*, *LabelEncoder*), clustering (*KMeans*), dimensionality reduction (*PCA*), and validation (*silhouette_score*).

Plotly Express is used for interactive charts within Streamlit. Seaborn and Matplotlib power the static feature correlation heatmap.

The Streamlit application uses `st.cache_data` on the `run_pipeline` function so that repeated interaction with the dashboard (tab switching, user selection) does not trigger recomputation. The pipeline accepts any valid EduPro-format Excel workbook, making the system generalizable to other e-learning platforms that conform to the three-sheet schema.

A custom CSS theme implementing Neo-Tech Minimalism aesthetics is injected via `st.markdown` with `unsafe_allow_html=True`, producing a dark glassmorphism interface suited to a modern data dashboard context. The sidebar provides dataset upload functionality, and pipeline status is communicated via `st.spinner` and `st.success` notifications.

VI. CONCLUSION

This paper presented a complete Student Segmentation and Personalized Course Recommendation System built on K-Means clustering, PCA, and behavioral feature engineering. By engineering nine meaningful features from transactional e-learning data and clustering learners into four interpretable personas, the system provides actionable insights for platform administrators and personalized course suggestions for individual learners.

The Streamlit-based interactive dashboard enables non-technical stakeholders to upload their data, explore cluster visualizations, inspect individual learner histories, and receive real-time recommendations without any programming knowledge. The Elbow Curve and Silhouette Score metrics provide transparent model validation.

Future work will explore deep learning-based embedding models for richer learner representations, temporal modeling to capture evolving learning behavior over time, and A/B testing of recommendation quality using platform engagement metrics. Integration of explicit content-based signals (course description embeddings, prerequisite graphs) into the recommendation engine is also planned.

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